

Milestone 1 – Module 2: Data Cleaning and Structuring

1. Methodologies

Module 2 focuses on transforming the raw scraped dataset into a **clean, standardized, and analytics-ready format** suitable for visualization.

1.1 Input Data Validation

- The raw dataset (`military_raw_data.csv`) generated in Module 1 was loaded into a Pandas DataFrame.
- Initial inspection was performed using:
 - `DataFrame.info()` for schema and data type validation
 - `isnull().sum()` for missing value detection
- Verified **one-row-per-country integrity** using `Country_Full_Name` as the primary identifier.

1.2 Identifier Preservation

The following columns were preserved **without modification** to ensure reliable joins and dashboard filtering:

- `Country_Full_Name`
- `Country_Short_Name`

These identifiers act as **primary keys** across analytics layers.

1.3 Numeric Standardization Pipeline

A reusable cleaning function was implemented to normalize all metric values.

The cleaning logic:

- Removes:
 - Commas, currency symbols, percentage signs
 - Line breaks and extra whitespace
- Strips unit labels:
 - USD, km, MT, BBL, Cu.M, Cum
- Converts cleaned strings into numeric values

This ensured **uniform numeric representation** across all metrics.

1.4 Data Type Conversion Rules

- **Power_Index**
 - Converted to float
- **All other metrics**
 - Converted to nullable integer type (Int64)
 - Allows safe handling of missing values without data loss

This approach guarantees:

- Accurate aggregations
- BI-tool compatibility
- Robust statistical analysis

1.5 Missing Value Handling

- Missing values were explicitly identified post-conversion.
- Logical imputation was applied only where meaningful:
 - Example: `total_naval_fleet_tonnage_mt` filled with 0 where absent
- Remaining nulls were retained to avoid misleading assumptions.

1.6 Final Dataset Generation

- The cleaned dataset was consolidated into a single DataFrame
- Exported as a CSV file for downstream analytics and dashboards

2. Libraries Used

Library	Purpose
pandas	Data manipulation, cleaning, type conversion
re (regex)	Removal of units, symbols, and formatting noise
numpy (implicit via pandas)	Numeric handling and NaN support

3. Output

3.1 Generated Dataset

File Name	Description
<code>military_cleaned.csv</code>	Fully cleaned, numeric, analysis-ready dataset

3.2 Dataset Characteristics

Attribute	Value
Countries Covered	145
Total Metrics	54
Categories	8
Rows	145
Columns	50+
Data Type	Fully numeric (except identifiers)

3.3 Final Dataset Structure (Sample)

Country_Full_Name	Power_Index	total_population	fighter_aircraft	tanks	defense_budget_usd	
India	0.1023	1409128296	576	4614	81000000000	

3.4 Intended Usage

The cleaned dataset is optimized for:

- Power BI / Tableau dashboards
- KPI derivation
- Country-wise military comparison
- Ranking and normalization models

Module 2 Outcome

- Raw data standardized into numeric format
- Missing values safely handled
- One-row-per-country integrity maintained
- BI-ready dataset generated
- Ready for KPI engineering and visualization